

Seventh Information Systems International Conference (ISICO 2023)

Improving Warehouse Layout Effectiveness and Process Picking Efficiency with the Discrete Event System Simulation Approach

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Abstract

Field conditions show process picking inefficiencies caused by the layout and arrangement of products in the warehouse that are not optimal. The objective of this research is to estimate the actual system performance and propose a new design so that the system becomes more effective and efficient. The research methodology combines Discrete-Event System (DES) Simulation to estimate the performance of the system and Double ABC Analysis to determine the re-arrangement of the item's position based on the pickup frequency and outer volume of the item. The statistical results identified the moving time has a significant difference between the actual and proposed layout. This study concludes that the re-arrangement of items can reduce the picking time of operators in the operational warehouse process.

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Peer-review under responsibility of the scientific committee of the Seventh Information Systems International Conference

Keywords: Discrete-Event System; Simulation; Warehouse; Layout

1. Introduction

Sales orders that occur every day can vary according to customer requests. The increase in the number and frequency of sales orders resulted in high lead time processes, which include order administration, picking, checking, packing, and loading goods. The picking process is a process that takes quite a long time compared to the checking process, packing process, and loading process. The actual condition with a large number of goods, irregular placement of goods, and the increasing number of stock items in the warehouse to meet customer needs will impact the length of the picking process.

Warehouse capacity that becomes overloaded and the position of piled up goods results in a lack of effectiveness in the movement of goods. Goods that have just entered (Inbound) will be placed in the front position close to the Staging Area so that the First In First Out (FIFO) process does not run well, especially in the oil goods storage section.

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The irregular placement of goods results in both the retrieval process for parts and oil being less efficient. Problems also occur with expired goods such as tires for motorcycles that have a large number of, overloading the warehouse capacity. The placement of these expired tires has taken up the racking capacity and forklift lane.

The FIFO process and the process of picking up goods based on fast-moving did not work well for oil and parts due to the irregular position of the goods. Improvements are needed to rearrange the position of goods in the warehouse with the approach of picking strategy and Double ABC Analysis as a helper to carry out a picking strategy based on the number of picking frequencies and volumes of goods to reduce the time of picking goods, First In First Out (FIFO) as an action to the circulation of the first incoming goods can leave the warehouse and the last incoming goods will be used as stock items, and from the results of this improvement will result in a reduction in picking process time.

Arena Simulation as one of the DES simulations is used to simulate the processing time of the warehouse lead time to measure the effectiveness and efficiency of the warehousing process. Reducing the time of the picking process is one of the quality management topics because it reduces time without lowering the quality of the process. The circulation zone is the area that can be passed without disturbing others, the individual zone is the area where one can pass between others, the no-touch zone is the area where one cannot pass between other people, and the touch zone is the area where one cannot pass through other people without disturbing the other person [1]. Facilities with well-organized material flows have closer or shorter transportation routes [2]. Transportation is always expensive in terms of space, machinery, and energy, but the high cost of transportation in the warehouse does not add value or value to the object being moved. In many existing facilities, the reduction and simplification of material movement can significantly lower material transportation costs. More important from a human point of view is the possibility of reducing the risk of overwork and injury to workers by redesigning, upgrading, or eliminating transportation routes.

According to Banks, Carson II, Nelson, & Nicol, simulation is an imitation or imitation of the operation of a process or system in real conditions from time to time [3], by using computer assistance, including the creation of data and the artificial history of a system. The ABC analysis method of stock movements and stocks held (on hand) must be carried out [4]. Understanding the ABC classification begins with understanding Pareto's Law or the 80/20 rule. It states that about 80 percent of the effects come from 20 percent of the causes. Because it is to concentrate time and resources on the 20 percent that are important or "the vital few". As a result, organizations can implement appropriate control policies by identifying the most effective criteria for classifying their inventory. Flexibility is an attribute of ABC analysis, and it allows one to prioritize tasks or items according to their greatest interests and benefits or briefly many different topics that can be run.

This research is considered a one-way or single-factor analysis of variance (ANOVA) because only one factor is examined, namely the placement of goods based on Double ABC Analysis [5]. It is implied that the research was conducted with the ultimate goal of testing the hypothesis of whether there was a significant change in the design results. Testing the hypothesis, the error model is assumed to be a normally distributed random variable and is independent. The F test in regression functions as a simultaneous test and is used to determine that all independent variables simultaneously have a significant influence on the dependent variable.

2. Method

The research methodology is a problem-solving framework that describes the stages of problem-solving briefly and their explanations. In general, the research methodology is structured to achieve the research objectives that have been set. The research problem activity is identified as related to the warehouse layout, the FIFO process of goods in the warehouse, and the process of picking goods, so the information learned in the literature study must be related to the Warehouse Management System and System Simulation. Warehousing plays a role in storing and exchanging goods for centuries [6]. Long-term storage to provide products for future consumption has become a warehousing utility both then and now. The FIFO system is a goods storage system; namely, the goods that enter first are issued first [7]. So the release of goods sequentially or according to chronology. This system is usually used for less durable goods.

Simulation is very important because it can prevent catastrophic failures in real systems due to the impact of new changes, procedures, information flow, and others [8]. Inspections can be carried out without disturbing the smooth running of the real system. A simulation model is developed to study the workings of a system that evolves, a fully developed and validated model can answer various questions about the real system. Analysis of simulation output can

be divided into qualitative and statistical analysis [9]. Some of the simulation studies are concerned with the estimation of the source system performance measures. Research about container terminal logistics optimization has several discussions, including, the yard sub-system's equipment management and space assignment, the handling equipment's performance, and the current actual facility housekeeping practices [10]. Shabayek and Yeung developed a simulation model for cost analysis of the container terminals, for the use of the container vessels' forecasted average inter-arrival time, and the estimation of the terminal operators' performance improvement when their handling capacities are varied [11]. Ahmad et al. proved that the researcher and the business provider could derive a solution for a non-stationary time-dependent queueing system improvement with limited resources using TRIZ and ARENA simulation to apply a systematic problem-solving tool [12].

This research collected interarrival processing time, picking time, check and packing time, as well as loading time. The next stage was simulating the model of the existing system, including verifying and validating the model. The validated model is simulated, and the output results data are recorded based on the specified criteria, namely the average waiting time for entities, the average number of waiting entities, and the average operator movement time when processing entities. The condition of the warehouse performance system with actual conditions is used to measure the performance of the actual condition after a simulation test is carried out using the data that has been collected—using the Goodness of Fit Test to determine the probability distribution for each process. According to Kelton, Sadowski, & Sadowski, this tool is used to determine the probability distribution function of the input data [13]. It can also be used to match specific functions of a data file distribution and compare distribution functions or display the effects of parameter changes for the same distribution. The input analyzer displays the random input data, which can then be analyzed using the fitting distribution software feature to find the distribution form that fits the data.

According to Chen and Lee, the actual condition system model has been verified by the company and experts [14]. Verification refers to checking the model to see if the conceptual model is accurately coded into a computer program. At the same time, validation involves checking the model to see if the model adequately represents the real system so that the simulation objectives can be achieved. The simulation system model already represents the Bandung warehouse outbound process. Figure 1 shows the system model of the existing conditions.

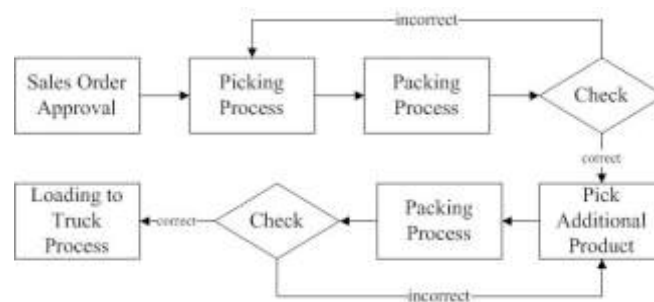


Fig. 1. Conceptual Model

The t-test is calculated by calculating the sample mean, standard deviation, and t-test statistics for n replications using the output number out. After testing the t-test data, it is continued by calculating the half-length criterion which can be assumed as the maximum limit for the error criterion. Determination of the acceptance of the simulation model is in the valid category and requires model improvement depending on the results of the calculation of the best-case error and worst-case error in the confidence interval. Because the average value of the actual condition is known and the WCE value is greater than the Error Criterion (ϵ) which is five, additional replication is needed.

The calculation of the output after additional replication is needed to measure the system's performance in actual conditions. The confidence interval and prediction interval are calculated if the average, standard deviation, and half-width have been obtained. The confidence interval is used to determine the average range of values from the simulation data. The Prediction Interval is designed to be quite broad compared to the confidence interval, containing the actual average cycle time on a particular day, with the highest probability. After calculating the point estimator, then we can calculate the confidence interval specifically for the quantiles.

There are several data collected to implement the double ABC analysis method to determine 20% of the capacity of the goods in the warehouse, which has 80% of the total frequency of collection and volume of goods. This research also collected distance data, racking and pallet size data, warehouse layout, inventory data, and time-motion data. The re-layout of inventories in the warehouse is designed according to double ABC analysis. According to Elizandro and Taha, using measurement experiments, such as an experimental design, can help minimize some uncertainties in performance measures [15]. The analyst will be able to evaluate an environment using experimental design techniques in which changes in the system's workload during a measurement period may introduce bias and performance measures variability. The analyst will also be able to assess the performance measures effects of changes to the system. After tabulating the actual and experimental simulation results, the next step is to perform statistical tests to determine the results of the hypothesis.

This study analyzes the difference in warehouse system performance between the initial conditions and the proposed layout based on the results of double ABC analysis. There is also research that uses discrete-event system simulation to analyze warehouse system performance by comparing several scenarios, with differences in the number and type of operators who perform the picking process [16]. Integration between discrete-event system simulation and other approaches is widely applied in optimizing warehouse systems. One of them is by adapting advanced computational solutions that integrate metaheuristic methods with simulations to get the most optimal storage location assignments which are considered to be less expensive than automated racks and robots [17]. Other researchers have integrated discrete-event system simulation with probabilistic simulation and optimization at a large retailer warehouse to improve picking performance [18]. Other studies have integrated object-oriented discrete-event system simulation with the random neighborhood search method for warehouse worker allocation, which is expected to contribute to workforce planning in a warehouse [19]. There is also research that does not combine discrete-event system simulation with other approaches, but uses it for complex systems where order picking in the warehouse is carried out simultaneously by human operators and autonomous robots [20].

3. Results and Discussion

A simulation test of the actual system conditions was carried out based on the actual condition of the warehouse layout, and the results of the validity calculations for the actual condition simulation test can be seen in Table 1. This research has determined the warehousing system performance criteria to measure the efficiency of performance, as shown in Table 2, and the average results from the simulation study of the actual condition system are 121 replications based on the criteria shown in Table 3.

An example of the results of the calculation of the output analysis for the criteria for the average queue time for the picking process in actual conditions can be seen in Table 4. The results of measuring the actual warehouse system performance for the average time waiting for the picking parts process are estimated that there is an error or error in the assumption of the average waiting time for the picking parts process for ± 1.51 minutes, with the probability that it occurs is 43% for the upper limit, then the lower limit with the probability of occurring is 43%.

Table 1. Actual Condition Validity Test

Simulation Mean	203	ϵ	5
Actual Mean	201	CI^+	210
Std. Deviation	28	CI^-	197
df	73	Best Case Error	9
T_{count}	0.75	Worst Case Error	4

Table 2. Actual Condition Validity Test

Picking Queue Time	Average waiting time for the n sales order for picking.
Picking Queue Mean	Average number of sales orders waiting for the picking process
Moving Time	Average operator operating time



Fig. 2. (a) Current Layout; (b) Propose Layout.

It is estimated that 95% that there is a risk that the average waiting time for the picking parts process in the two days at the end of the following month will have an additional or reduced time of ± 16.67 minutes with a probability of occurrence of 2% for the upper limit, then the lower limit with the probability of occurring is equal to 21%. The point estimator has a value of 6.68 minutes based on the 0.75th quantile of a lot of data, namely 121 replications. A special confidence interval for quantiles in the parts category produces an upper limit $\hat{\theta}_u = 9.37$ with a probability of 37% occurring, then a lower limit $\hat{\theta}_l = 5.68$ with a probability of occurring at 45%. The confidence interval of average data gained information about the presence of outliers, while the confidence interval quantiles are calculated to identify the absence of outliers that interfere with the calculation of opportunities.

According to the performance analysis, the proposed layout was developed to improve the system. Figure 2 (a) shows the actual condition of the warehouse layout, where the preparation of items is only based on categories and several subcategories. Figure 2(b) shows the condition of the warehouse layout design where the preparation of goods is based on the frequency of taking, the volume of cartons of goods, and arranged by category and several

subcategories to make it easier for operators to find goods after rearranging goods. Based on the warehouse layout for the design conditions, a simulation test of the condition of the design system was carried out, and the results of the validity calculations for the simulation test of the design conditions can be seen in Table 5.

Table 3. Results Average Performance System Performance Actual Condition.

	Actual					Proposed				
	A	B	C	D	E	A	B	C	D	E
Average	6.59	1.40	0.41	0.25	1.58	8.02	1.40	0.47	0.23	1.11
Average Max	64.38	4.53	3.57	1.03	1.83	146.15	5.18	10.5	0.59	1.27
Average Min	0.29	0.19	0.01	0.02	1.36	0.31	0.32	0.01	0.04	0.96
A :	Picking Parts Queue Time (min)					D :	Picking Oil Queue Mean (pcs)			
B :	Picking Oil Queue Time (min)					E :	Moving Time Parts (min)			
C :	Picking Parts Queue Mean (pcs)					F :	Moving Time Oil (min)			

Table 4. Output Analysis of Picking Queue Time Actual and Proposed Design.

Picking Parts Queue Time (min)	Actual	Proposed	Picking Oil Queue Time (min)	Actual	Proposed
Quantiles	0.75	0.75	Quantiles	0.75	0.75
Half-width	1.51	2.91	Half-width	0.14	0.16
Std. Dev. quantiles	8.385	15.13	Std. Dev. quantiles	0.778	0.83
$\bar{Y} \pm H$	6.59 ± 1.51	8.02 ± 2.91	$\bar{Y} \pm H$	1.40 ± 0.14	1.40 ± 0.16
CI ⁺	8.10	10.94	CI ⁺	1.54	1.55
CI ⁻	5.08	5.11	CI ⁻	1.26	1.24
$\Pr\{\bar{Y}_t \geq 8.10\}$	43%	42%	$\Pr\{\bar{Y}_t \geq 1.54\}$	43%	42%
$\Pr\{\bar{Y}_t \leq 5.08\}$	43%	42%	$\Pr\{\bar{Y}_t \leq 1.26\}$	43%	42%
PI ⁺	23.26	38.17	PI ⁺	2.95	3.04
PI ⁻	$-10.08 \approx 0$	$-22.13 \approx 0$	PI ⁻	$-0.14 \approx 0$	$-0.25 \approx 0$
$\Pr\{\bar{Y}_t \geq 23.26\}$	2%	2%	$\Pr\{\bar{Y}_t \geq 2.95\}$	2%	2%
$\Pr\{\bar{Y}_t = 0.00\}$	21%	30%	$\Pr\{\bar{Y}_t = 0.00\}$	35%	46%
Point Estimator	6.68	8.16	Point Estimator	1.72	1.67
CI ⁺	0.83	0.83	CI ⁺	0.83	0.83
$101^{\text{th}}(\hat{\theta}_u)$	9.37	11.42	$101^{\text{th}}(\hat{\theta}_u)$	1.99	1.88
CI ⁻	0.67	0.67	CI ⁻	0.67	0.67
$81^{\text{th}}(\hat{\theta}_u)$	5.68	6.42	$81^{\text{th}}(\hat{\theta}_u)$	1.43	1.47
$\Pr\{\bar{Y}_t \geq 9.37\}$	37%	41%	$\Pr\{\bar{Y}_t \geq 1.99\}$	36%	40%
$\Pr\{\bar{Y}_t \leq 5.68\}$	45%	46%	$\Pr\{\bar{Y}_t \leq 1.43\}$	36%	41%

Table 5. Design Result Condition Validity Test.

Simulation Mean	198	€	5
Actual Mean	201	CI+	210
Std. Dev.	26	CI-	197
Df	73	Best Case Error	9
T _{count}	0.84	Worst Case Error	4
T _{table}	1.99	Replication	106

The average results from the simulation study of the actual condition system with 106 replications based on the criteria that the company has determined are in Table 3. An example of the calculation results of the output analysis for the criteria for the average queue time of the picking process for the design conditions can be seen in Table 4.

The results of measuring the actual warehouse system performance for the average time waiting for the picking parts process are estimated that there is an error or error in the assumption of the average time waiting for the picking parts process for ± 2.91 minutes, with the probability that it occurs is 42% for the upper limit, then the lower limit with the probability of occurring is 42%.

It is estimated that 95% that there is a risk that the average waiting time for the picking parts process in the two days at the end of the following month will have an additional or reduced time of ± 30.05 minutes with a probability of occurring 2% for the upper limit, then the lower limit with the probability of occurring is equal to 30%. The point estimator has a value of 8.16 minutes which is obtained based on the 0.75th quantile of a lot of data, namely 106 replications. Based on a special confidence interval for quantiles in the parts category, which produces an upper limit $\hat{\theta}_u = 11.42$ with a probability of 41% occurring, then a lower limit $\hat{\theta}_l = 6.42$ with a probability of 46% occurring.

Based on the results of the statistical test using One-Way ANOVA in Table 6 and Table 7, it is found that only the Moving Time criteria have changed significantly because at the time of rearranging the goods that can be measured and calculated for the design results in only the distance and travel time of the operator when carrying out picking up goods in the racking area.

Table 6. ANOVA Test Table for 'Parts' Moving Time Category.

Sources	Sum Square	DoF	Mean Square	F _{count}	P _{value}	F _{table}
Movement Time	12.77	1	12.77	2,563.39	0.00	3.88
Error	1.12	225	0.00498	F count > F crit		
Total	13.89	226	Reject H0			

Table 7. ANOVA Test Table for 'Oil' Moving Time Category.

Sources	Sum Square	DoF	Mean Square	F _{count}	P _{value}	F _{table}
Movement Time	53.79	1	53.79	77,153.98	0.00	3.88
Error	0.16	225	0.00070	F count > F crit		
Total	53.95	226	Reject H0			

ANOVA test states $F_{\text{count}} < F_{\text{table}}$ for the average picking queue time and the average picking queue with parts and oil categories. So it can be concluded that the results state Do not Reject H_0 , which means that the results of the hypothesis test between the actual conditions and the results of the design state that there is no significant change. The reason there is no significant change and experiencing additional time and the number of queues from the actual conditions for the performance criteria of the average picking queue time and the average picking queue is since the processing time for picking goods at the racking station for the design conditions is unknown and causes bottleneck indications.

Time reduction was successfully carried out for the moving time criteria, and an ANOVA test was carried out stating $F_{\text{count}} < F_{\text{table}}$ for the moving time criteria with parts and oil categories. So it can be concluded that the results stated Reject H_0 , which means that the results of the hypothesis test between the actual conditions and the design results stated that there was a significant change. The reason there is a significant change and a reduction in travel time from the actual conditions for the moving time performance criteria is that the travel time for picking up goods at the design warehouse can be calculated and measured based on the distance traveled and the average walking speed of humans and the average speed of the forklift operating.

4. Conclusion

The results of measuring the actual system performance of the Bandung warehouse concluded that it is not optimal enough. For example, the average waiting time for the picking parts process has a time of 6.59 minutes which is

estimated that there is an error in the assumption of the average waiting time for the picking parts process for ± 1.51 minutes with the probability occurring was 43% for each upper-lower limit. It is estimated that 95% that there is a risk that the average waiting time for the picking parts process in the two days at the end of the following month will have an additional or reduced time of ± 16.67 minutes with the probability of occurring is 2% for the upper limit and 21% for the lower limit.

The point estimator has a value of 6.68 minutes based on the 0.75th quantile of 121 replications. Based on the confidence interval for the specific quantiles in the parts category, which results in an upper limit of 9.37 minutes with a probability of occurrence of 37%, then a lower limit of 5.68 minutes with an occurrence probability of 45%. The results of the rearrangement of goods based on the frequency of taking and the volume of outer goods using double ABC analysis optimize operator operating time to be more efficient with the results of hypothesis testing H_0 being rejected, namely $F_{\text{count}} \geq F_{\text{table}}$ because the travel time for picking up goods at the design warehouse can be calculated or measured based on the distance traveled and the average human walking speed and the average forklift operating speed.

This research contributes as a best practice in business operations in grouping goods using the double ABC analysis method and arranging them following the structuring pattern offered in this study. This research methodology contributes to academics in the form of a framework in determining the optimal layout and proving it using statistical analysis based on the resulting data from the discrete-event system simulation. For future research, it is recommended to include stock variables in the simulation test to cover the process of picking goods based on FIFO specifically, and experimental warehouse conditions are applied so that the picking process time can be measured to show significant changes.

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